

National University of Computer and Emerging Sciences, Lahore Campus



Course: Analog and Digital Communications
 Program: BS(Electrical Engineering)
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 Page(s): 3
 Roll No: 115-4417

Instruction/Notes: Attempt all parts of a question together.
 Fourier Transforms Table is provided

[15]

Question # 1 [CLO 1]:

A low-pass filter (figure 1) transfer function $H(\omega)$ is given by

$$H(\omega) = \begin{cases} (1 + k \cos \omega T) e^{-j5\omega} & |\omega| < 2\pi B \\ 0 & |\omega| > 2\pi B \end{cases}$$

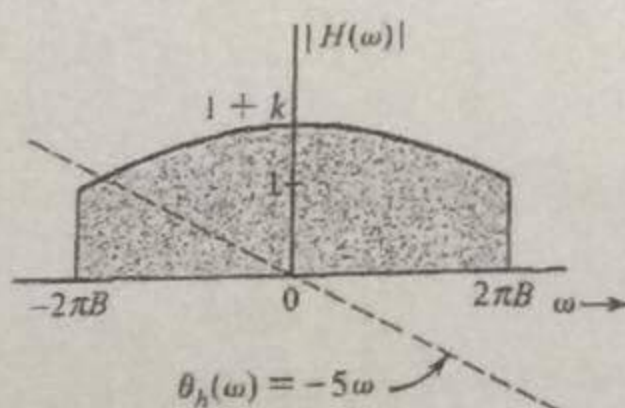


Figure 1

- ✓ Is this a distortion-less system if yes explain, if no then identify the type of distortion introduced by the system.
- ✓ A pulse $g(t)$ band-limited to B Hz is applied at the input of this filter. Find the output $y(t)$ in terms of $g(t)$ and describe the effects of channel on input signal.

Question # 2 [CLO 2]:

[15]

A random binary pulse train $g(t)$ is shown in figure 2. The pulse width is T_w , and one binary digit is transmitted every T_w seconds. A binary 1 is transmitted by the positive pulse, and a binary 0 is transmitted by the negative pulse. The two symbols are equally likely and occur randomly.

- Find the autocorrelation function $\mathcal{R}_g(\tau)$ and plot it.
- Find the power spectral density $S_g(\omega)$, plot it and comment on the spectral characteristics of the pulse train.
- Estimate the essential bandwidth of the signal.

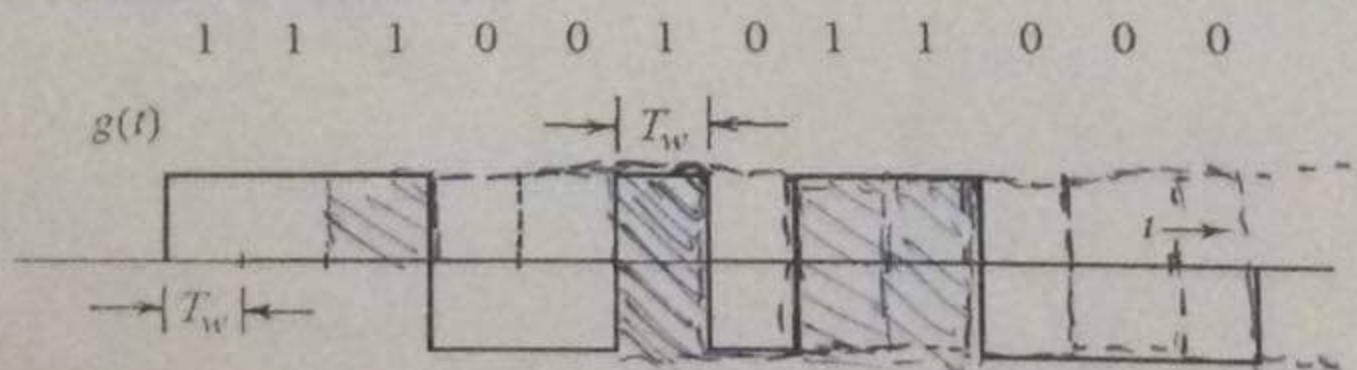


Figure 2

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Essential Bandwidth = ?

Question #3 [CLO 3]

An amplitude modulator is a device that accepts two inputs $x(t)$ and carrier $c(t)$, and generates the product of the two signals on the output $y(t) = x(t)c(t)$. In practice, an analog multiplier may be difficult to build. Therefore practical systems often use a nonlinear processing element and additional processing to produce the same effect as multiplication. Consider the system in figure 3 that uses a squaring nonlinearity $z(t) = w^2(t)$. Carrier frequency is $\omega_c = 10\pi \text{ rad/s}$

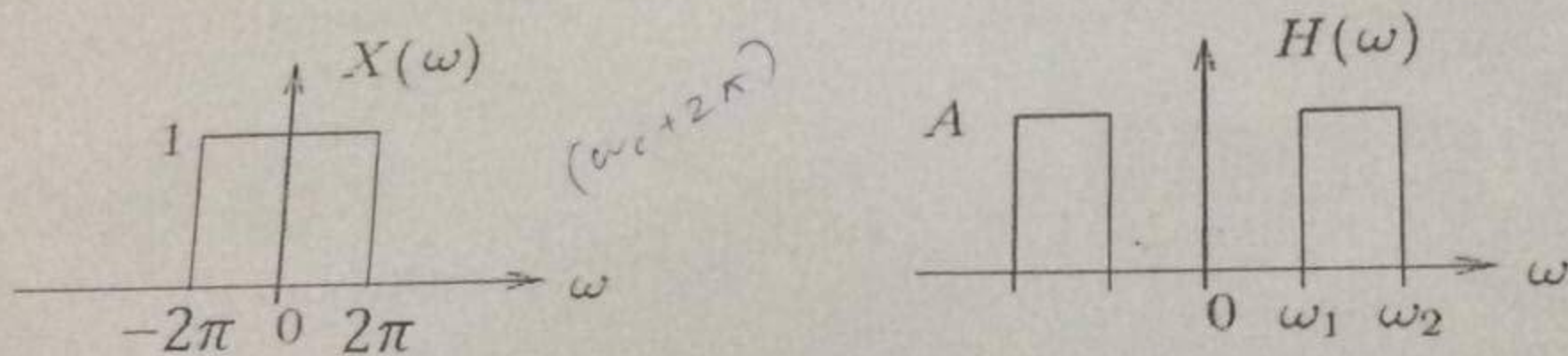
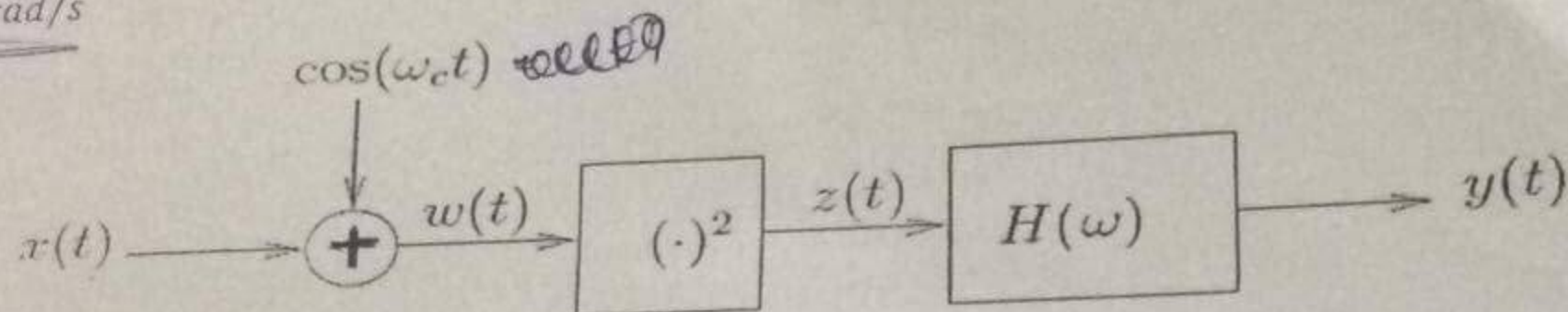


Figure 3

The spectrum of $x(t)$ and $H(\omega)$ are also shown. The filter $H(\omega)$ is an Ideal Band-pass filter with Impulse response:

$$H(\omega) = \begin{cases} 1 & 8\pi \leq \omega \leq 12\pi \\ 0 & \text{otherwise} \end{cases}$$

- Write mathematical expressions for $w(t)$, $z(t)$ and $y(t)$ [Simplify using trigonometric identities where required].
- Write expressions for $W(\omega)$, $Z(\omega)$ and $Y(\omega)$ also sketch them.
- Determine the bandwidth of modulated signal $y(t)$.